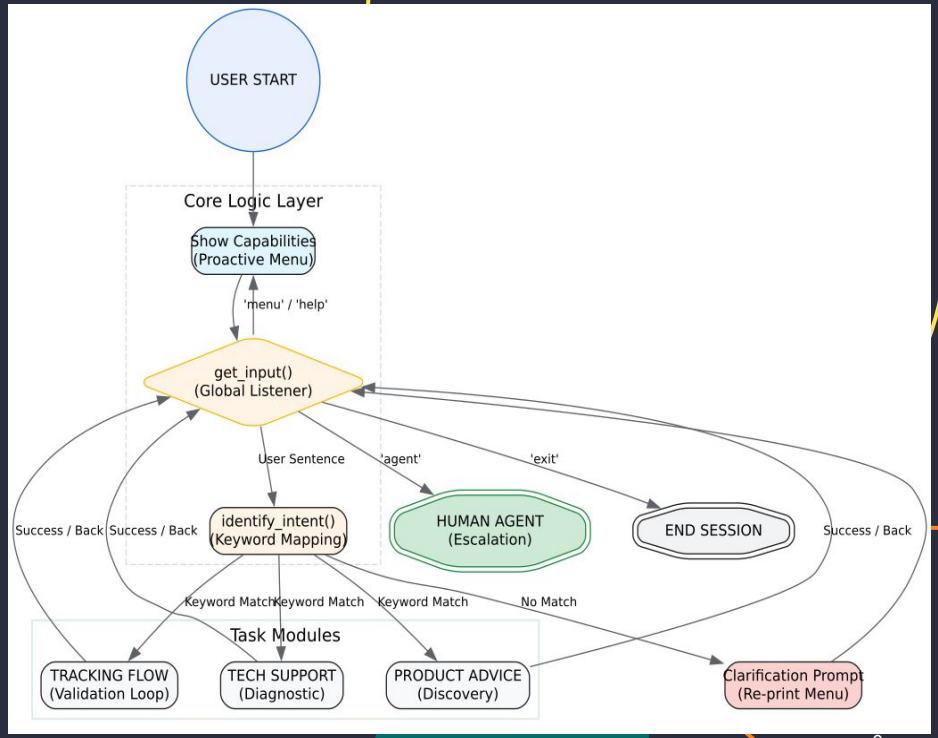
Customer Service Chatbot

Design Choices

Discoverability over Mystery: Instead of a blank prompt, the bot starts with a `show_capabilities()` menu. This eliminates the "What do I type?" anxiety.
Simulated Empathy: Implemented a `bot_type` function with a variable delay. This mimics human typing speeds, making the interaction feel less robotic and easier to follow.
Global Command Listener: Designed a central input wrapper so that "MAGIC WORDS" like **MENU**, **HELP**, and **AGENT** work anywhere in the flow, giving the user a constant "Escape Hatch."
Binary & Numeric Constraints: Chose specific data formats (8-digit IDs, Yes/No prompts) to minimize the "Surface Area" for user errors.



Technical Implementation

Intent Recognition Engine: Used a keyword-clustering algorithm to map natural language phrases (e.g., "Where is my stuff?") to specific logic functions (`tracking`, `support`).
State Management: The bot uses a modular function-based design. Each scenario (Tracking, Sales, Support) is a self-contained "State" that can return the user to the Main Menu or escalate to an agent.
Input Validation Layer:

- **Type Checking:** Ensuring IDs are numeric.
- **Length Validation:** Ensuring IDs are exactly 8 digits.
- **Normalization:** Using `.lower()` and `.strip()` to handle casing and accidental spaces.

Challenges Faced

Handling Ambiguity: One major challenge was when a user provides "Multi-Intent" input (e.g., "I want to buy a laptop because mine is broken").

Infinite Loops: Users often get stuck entering the wrong info repeatedly.

The "Blank Screen" Problem: Initial testing showed users didn't know where to start.

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==============================================
        AI CUSTOMER SUPPORT TERMINAL
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Bot: Welcome! I am your automated assistant.

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💡 I CAN ASSIST YOU WITH:
1. TRACKING  - Check status of a lost package
2. PRODUCTS  - Get a recommendation for a device
3. SUPPORT   - Basic troubleshooting for errors

COMMANDS: Type 'AGENT' for a human or 'EXIT' to leave.
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How can I help you today? 
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Future Roadmap AI Evolution

Natural Language Understanding (NLU): Replace keyword mapping with a Large Language Model (LLM) to understand complex, conversational, and misspelled user inputs perfectly.
RAG Integration (Retrieval-Augmented Generation): * Connect the chatbot to a **Vector Database** containing all company manuals and real-time shipping data.

- *Benefit:* The bot provides "Ground Truth" answers based only on your data, preventing "hallucinations" while sounding natural and professional.

Contextual Memory: Implementing a "Conversation Buffer" so the bot remembers details from 10 messages ago (e.g., the user's name or a specific error code) without asking twice.
Proactive Personalization: Using historical data to predict why a user is calling (e.g., "I see your package was delayed this morning, are you calling about that?

Thank you